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India
SMART UTILITY
Week 2026

10 – 14 March 2026, New Delhi

Session : AI–ML–VR–AR and Robotics Applications for Utilities

Industry Use Cases of AR, VR, AI, aR (Assisted Reality) and Digital Twin in Energy Operations

Presented By

Keyur Bhalavat

CEO & Co-Founder Plutomen

Plutomen

THE FUTURE OF

Energy Distribution

Is Being Written Right Now.

Are you ready to lead — or be left behind?

Where Is the Energy Sector Heading?

Grid Modernization

Smart grids, SCADA upgrades, real-time monitoring of thousands of substations & feeders

Renewable Integration

Rapid DRE uptake demands agile field teams who can inspect & maintain at scale

Regulatory Compliance

CERC, State Regulators & safety mandates drive the need for audit-ready digital records

Ageing Workforce

Critical knowledge is walking out the door - organizations must capture & transfer expertize now

The Burning Challenges in the Field

Paper-Driven Operations

- Manual checklists & permit forms
- Delayed reporting & no traceability
- Inconsistent SOP execution

Knowledge Dependency

- Critical work relies on senior experts
- Expert unavailability = delays & errors
- No digital knowledge capture

Disconnected Field Teams

- No real-time guidance in the field
- Repeat visits & longer downtime
- Higher safety exposure

Compliance Gaps

- Audit prep is time-consuming
- PTW & maintenance records scattered
- Incident investigation slowed down

Virtual Reality (VR) — Transforming Utility Training

Immersive Safety Training

- Safe simulation of live-line & substation tasks
- 65% faster onboarding; 75% higher retention
- Practice emergency response without risk

Design & Operations Visualization

- Engineers explore 3D substation & transmission network models
- SCADA & control system simulation: practice load management safely
- Hazardous environment familiarization before field deployment

SCADA & Grid Simulation

- Interact with virtual SCADA panels & control rooms
- Collaborative VR for team grid planning exercises
- Virtual substation walkthroughs before deployment

Cost & Scale Advantages

- No physical infrastructure required for training
- Scale training across all geographies simultaneously
- RDSS-eligible CAPEX for Indian DISCOMs

AR & Assisted Reality (aR) — Empowering Field Operations

AR - Real-Time Field Guidance

- Live SCADA data overlaid on physical equipment
- Step-by-step repair instructions without paper
- LOTO & hazard zone alerts in the technician's view

aR - Hands-Free Expert at Your Side

- Remote expert sees exactly what technician sees
- Monocular display; full situational awareness retained
- Voice & gesture control; PPE-compatible rugged devices

AR in Inspections & Compliance

- Augmented inspections with real-time asset tagging
- Maintenance history & manuals accessed on-the-go
- Voice-controlled audit checklists with photo/video capture

aR in Power Sector - Field Impact

- SOPs & checklists in line-of-sight; voice-controlled task execution
- Instant access to specs, maintenance records & sensor data on-site
- PPE-compatible rugged devices with real-time safety warnings

Digital Twin — The Living Mirror of Your Grid

What is a Digital Twin?

- A real-time virtual replica of every physical asset
- Fed by IoT sensors & SCADA in real time
- Simulates loads, faults & failure scenarios

Grid Planning & Design Optimization

- Test layouts & strategies before capital investment
- Simulate DER & renewables integration on live grid
- Retrofit alignment & spatial planning from 3D overlays

Predictive Asset Management

- ML models forecast degradation before failures occur
- Condition-based maintenance replaces time-based schedules
- Investment prioritization aligned to real asset health

Technology Convergence at the Core

- ML + DT: Self-Learning Grid - adapts & predicts failures over time
- AR/VR + DT: Walk through your grid virtually; inspect internal parameters live
- Gen AI + DT: Converts manuals & logs into queryable, actionable knowledge

The Convergence: AI + AR + VR + Digital Twin

AI + Digital Twin

Self-Learning Grid

Continuous simulation + ML creates a grid that adapts and predicts failures before they strike

AR/VR + Digital Twin

Immersive Asset Intelligence

Walk through your digital twin in VR; inspect internal asset parameters live via AR overlay

AI + AR Field Guidance

AI-Powered Co-pilot

Gen AI turns manuals & logs into queryable knowledge; AR delivers answers in the technician's line of sight

The Full Convergence

Connected, Intelligent Utility

AI + AR + VR + Digital Twin form one ecosystem: predict, simulate, train, and resolve - all connected

Impact & Benefits for Energy Utilities

⬇40%	⬆95%	⬇60%	100%
Reduction in Downtime	First-Time Fix Rate	Expert Travel Cost	Audit-Ready Records

Safety	AR-guided LOTO, hazard zone alerts & SOP execution reduce field errors. aR safety warnings display in real time without obstructing view.
Real-Time Data Access	AR/aR devices surface live SCADA data, equipment history & sensor readings at the point of work — zero context switching.
Knowledge Retention	VR encodes expert knowledge digitally - new hires reach proficiency 65% faster with 75% higher retention

Recommendations for Indian Utilities

SHORT-TERM (0-12M)

Start with VR Training & AR Remote Assistance

Pilot VR modules for substation & HV equipment training. Deploy aR smart glasses for remote expert collaboration. Leverage RDSS to fund as CAPEX. Immediate ROI via reduced training costs & expert travel.

MEDIUM-TERM (1-3Y)

Digitize Field Ops with AR & Workflow Platforms

Deploy AR-guided digital workflows for inspections, PTW & audits. Integrate with SCADA & ERP.

Build audit-ready compliance records. Target PLI scheme KPIs: fault reduction & AT&C loss benchmarks.

LONG-TERM (3-5Y)

Build Digital Twin & Converged Intelligence Layer

Deploy substation & grid-level digital twins. Integrate ML for predictive maintenance.

Use AR/VR to interact with live digital twin data. Achieve the self-learning, connected utility of the future.

Plutomen

The Grid Is Changing.

Your Workforce Must Change With It.

Plutomen empowers energy & distribution companies to move from paper-based execution to connected, guided, and traceable field operations.

Keyur Bhalavat

CEO & Co-Founder, Plutomen

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